

# KARTHIK PASULA

JUNIOR BACKEND DEVELOPER | PYTHON · FASTAPI · POSTGRESQL · DOCKER

+91 8790518267 | [pasulakarthik03@gmail.com](mailto:pasulakarthik03@gmail.com) | [Pasulakarthik \(Karthik\)](#) | [Karthik Pasula](#) | [LinkedIn](#) Nandyal, India

## PROFESSIONAL SUMMARY

Backend developer with production experience building secure, containerized REST APIs using FastAPI, PostgreSQL, and Docker. Independently designed and deployed a JWT + RBAC authentication system with 15+ endpoints, achieving a 35% reduction in DB query response time through SQLAlchemy optimization. Comfortable across the full backend lifecycle — API design, database modeling, containerized deployment, and Linux environments. Actively seeking a junior backend or associate software engineer role to contribute to real-world systems.

## TECHNICAL SKILLS

Languages : Python, SQL, Bash Scripting

Backend : FastAPI, REST APIs, JWT Authentication, RBAC, Pydantic, argon2

Database : PostgreSQL, SQLAlchemy ORM, RDBMS, SQL Queries & Optimization

DevOps : Git, GitHub, Docker, Containerization Linux, VS Code, Render

Protocols : HTTP / HTTPS , JSON, OpenAPI / Swagger

Concepts : API Design & Integration, Authentication & Authorization, CRUD Operations, OOP

## PROJECTS

FastAPI Authentication API | Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, Docker, Pydantic, argon2, RBAC

- Designed and deployed a production-ready authentication system with 15+ RESTful endpoints following clean architecture and separation-of-concerns principles.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) for secure access management.
- Reduced DB query response time by ~35% via SQLAlchemy ORM optimization and indexed queries.
- Applied argon2 password hashing, secure session management, and Pydantic input validation to ensure data integrity.
- Structured modular backend architecture to ensure scalability and maintainability.

GitHub: [Pasulakarthik/new\\_api](#)

API Link: [Render - Application loading](#)

System Monitoring Tool | Bash, Linux

GitHub: [Pasulakarthik/Bash-Projects](#)

- Built a Bash script to monitor CPU, memory, and disk utilisation in real time with configurable alert thresholds.
- Implemented automated logging and threshold-breach notifications, reducing manual health-check effort by an estimated 80%.

Docker Deployment Automation | Bash, Docker

- Automated a complete Docker pipeline (image pull, container restart, port mapping, environment config) via Bash scripts, cutting manual deploy steps from ~15 commands to a single script execution.

Log Analyzer | Bash, Linux

- Parsed web-server logs to extract traffic insights, error statistics, and usage patterns — automating routine analysis that previously required manual grep pipelines.

Frontend Clones - Bigbasket & Flipkart | HTML, CSS

Webpage Link: [Bigbasket](#) [flipkart](#)

- Built responsive homepage clones of Flipkart and Bigbasket using HTML and CSS.
- Recreated navigation bars, product sections, banners and layouts with responsive design for multiple screen sizes.

## EDUCATION

Bachelor of Computer Science (B.Sc. Cs )

Sri Ramakrishna Degree College (Autonomous) – Nandyal, India

Expected June 2026

CGPA: 8.00 / 10

## CERTIFICATION

- Python – Reliance Foundation Skilling Academy (2025).
- Communication Skills – APSSDC (2024).